

논문 리뷰 260930

SuperPoint(CVPR 2018)

Self-Supervised Interest Point Detection and Description

<https://arxiv.org/abs/1712.07629>

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01 Intro

1. SuperPoint란?

- 학습 기반의 local feature extractor
- 전체 이미지를 보고 CNN모델이 pixel 단위에서 interest point를 계산.
- 관심점 검출의 반복성(repeatability)을 위해 Homographic Adaptation 도입
 - repeatability : 다른 조건(조명, 환경)에서도 같은 지점을 다시 찾아낼 수 있는가

Abstract

This paper presents a self-supervised framework for training interest point detectors and descriptors suitable for a large number of multiple-view geometry problems in computer vision. As opposed to patch-based neural networks, our fully-convolutional model operates on full-sized images and jointly computes pixel-level interest point locations and associated descriptors in one forward pass. We introduce Homographic Adaptation, a multi-scale, multi-homography approach for boosting interest point detection repeatability and performing cross-domain adaptation (e.g., synthetic-to-real). Our model, when trained on the MS-COCO generic image dataset using Homographic Adaptation, is able to repeatedly detect a much richer set of interest points than the initial pre-adapted deep model and any other traditional corner detector. The final system gives rise to state-of-the-art homography estimation results on HPatches when compared to LIFT, SIFT and ORB.

01 Intro

2. 기존의 방법들

- 고전방법들(Edge, blob)은 사람이 좋은 point의 기준을 정하는 알고리즘.

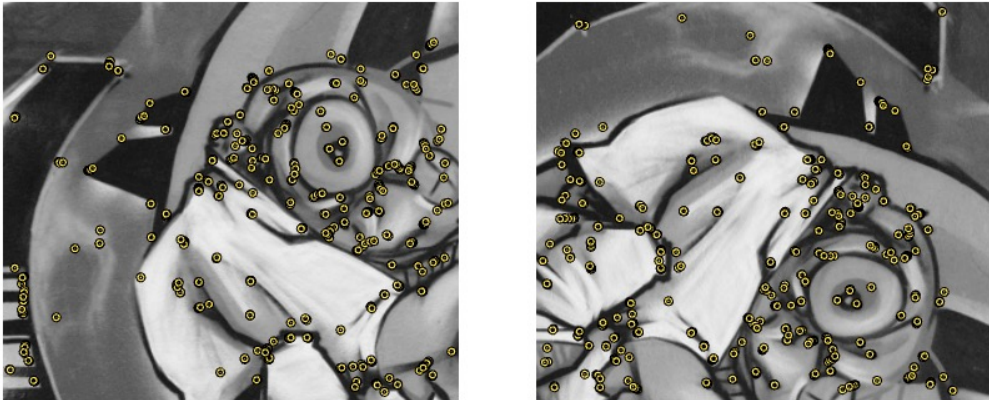


Fig. 3.2 Harris corners detected on rotated image examples.

< Harris : 미분 기반 corner >

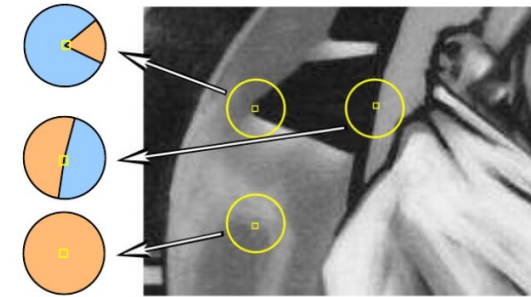


Fig. 3.3 SUSAN corners are detected by segmenting a circular neighborhood into “similar” (orange) and “dissimilar” (blue) regions. Corners are located where the relative area of the “similar” region (USAN) reaches a local minimum below a certain threshold.

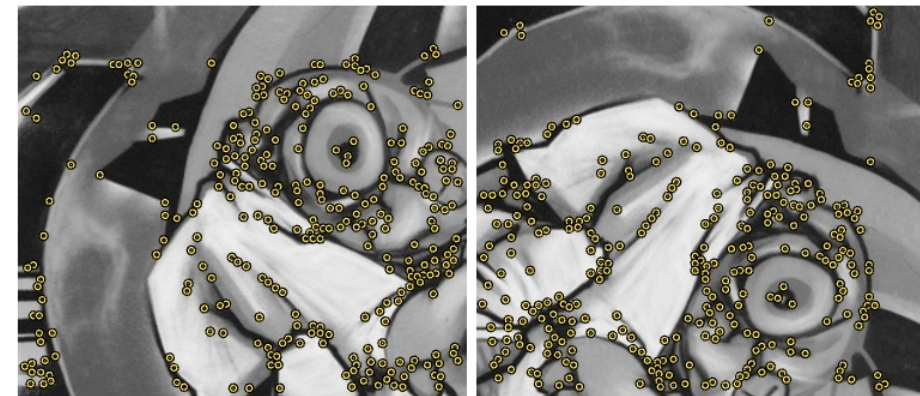


Fig. 3.4 SUSAN corners found for our example images.

< SUSAN : 면적 기반 corner >

01 Intro

2. 기존의 방법들

- 속도를 극한으로 끌어올린 고전 알고리즘

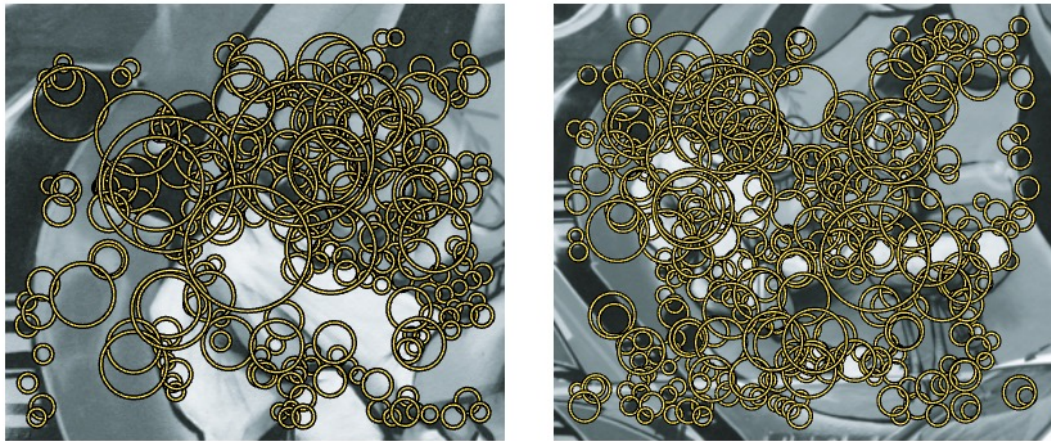


Fig. 6.6 Local features detected with the SURF-detector.

< SURF : Hessian 행렬 기반 blob >

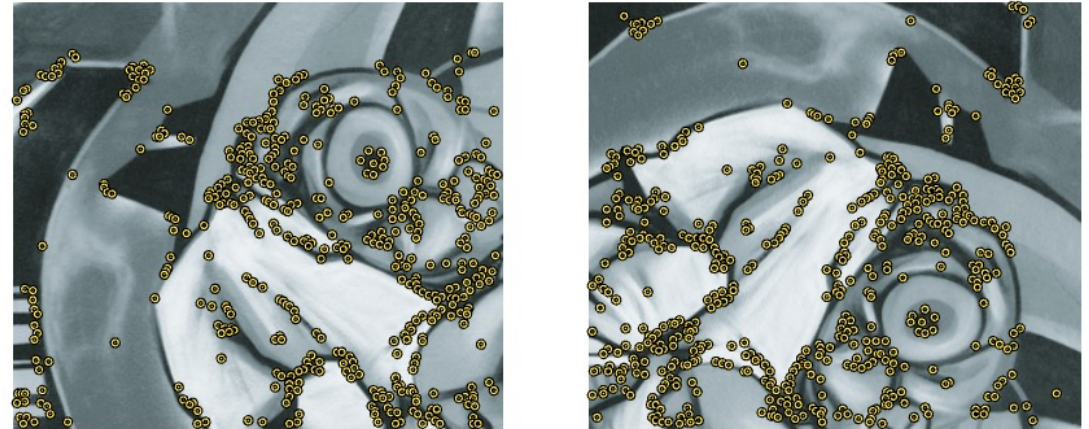


Fig. 6.8 Local features detected with the FAST detector.

< FAST : decision tree기반 corner >

01 Intro

3. (참고)이상적인 local feature detector의 속성

- local Invariant Feature Detectors: A survey(2008) <https://lvelho.impa.br/ip08/reading/features.pdf>

- Repeatability(반복성) : 다른 시점/각도에서도 동일하게 검출되어야
- Distinctiveness(변별력) : 검출된 점 주변의 패턴이 충분히 다양해야
- Locality(국소성) : 국소 영역만 보고 판단이 되어야(가려짐 occlusion)
- Quantity(충분한 개수) : 검출되는 점의 개수가 충분히 많아야
- Accuracy(정확도) : 위치/스케일/모양도 정확하게 맞아야
- Efficiency(효율성) : 실시간 응용에 쓸 수 있을만큼 빨라야

→ SuperPoint는 Repeatability에 집중한 논문

→ 학습을 시켜야하는데, Interest point는 Ground Truth가 없다.

→ Repeatability가 높은 점을 정답 데이터 없이 어떻게 학습으로 만들어낼까? 라는
데이터 생성 문제 -> Homographic Adaptation

Good features should have the following properties:

- *Repeatability*: Given two images of the same object or scene, taken under different viewing conditions, a high percentage of the features detected on the scene part visible in both images should be found in both images.
- *Distinctiveness/informativeness*: The intensity patterns underlying the detected features should show a lot of variation, such that features can be distinguished and matched.
- *Locality*: The features should be local, so as to reduce the probability of occlusion and to allow simple model approximations of the geometric and photometric deformations between two images taken under different viewing conditions (e.g., based on a local planarity assumption).
- *Quantity*: The number of detected features should be sufficiently large, such that a reasonable number of features are detected even on small objects. However, the optimal number of features depends on the application. Ideally, the number of detected features should be adaptable over a large range by a simple and intuitive threshold. The density of features should reflect the information content of the image to provide a compact image representation.
- *Accuracy*: The detected features should be accurately localized, both in image location, as with respect to scale and possibly shape.
- *Efficiency*: Preferably, the detection of features in a new image should allow for time-critical applications.

Repeatability, arguably the most important property of all, can be

02 SuperPoint

1. Main Contribution in detection

- 기존의 전통적인 방법들은 "좋은 점"의 기준을 사람이 수식으로 설계
- 좋은 Interest Point는 정의가 모호하다. → 데이터로 학습시키는 방향
- "객체탐지"와 같은 일반적인 supervised-learning이 불가능.

< 전체 학습 파이프라인 >

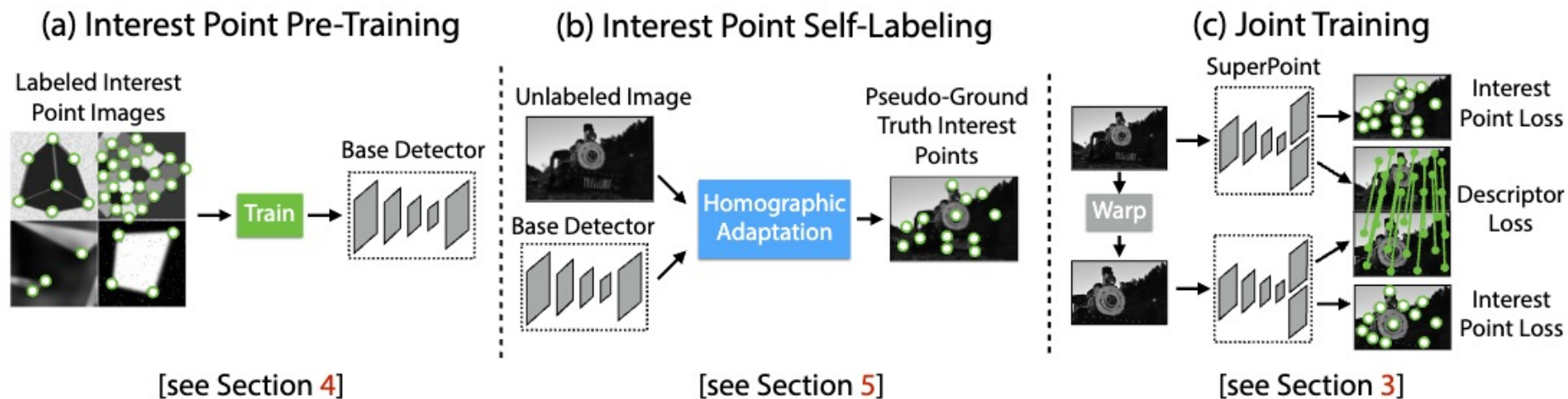


Figure 2. **Self-Supervised Training Overview.** In our self-supervised approach, we (a) pre-train an initial interest point detector on synthetic data and (b) apply a novel Homographic Adaptation procedure to automatically label images from a target, unlabeled domain. The generated labels are used to (c) train a fully-convolutional network that jointly extracts interest points and descriptors from an image.

(a)합성 데이터로 학습 (b)학습한 모델로 Homographic Adaptation 적용 -> 라벨 생성 (c)그 라벨로 학습 → SuperPoint

02 SuperPoint

2. SuperPoint Architecture

- 각 파이프라인 단계에서는 모두 하나의 아키텍처를 활용

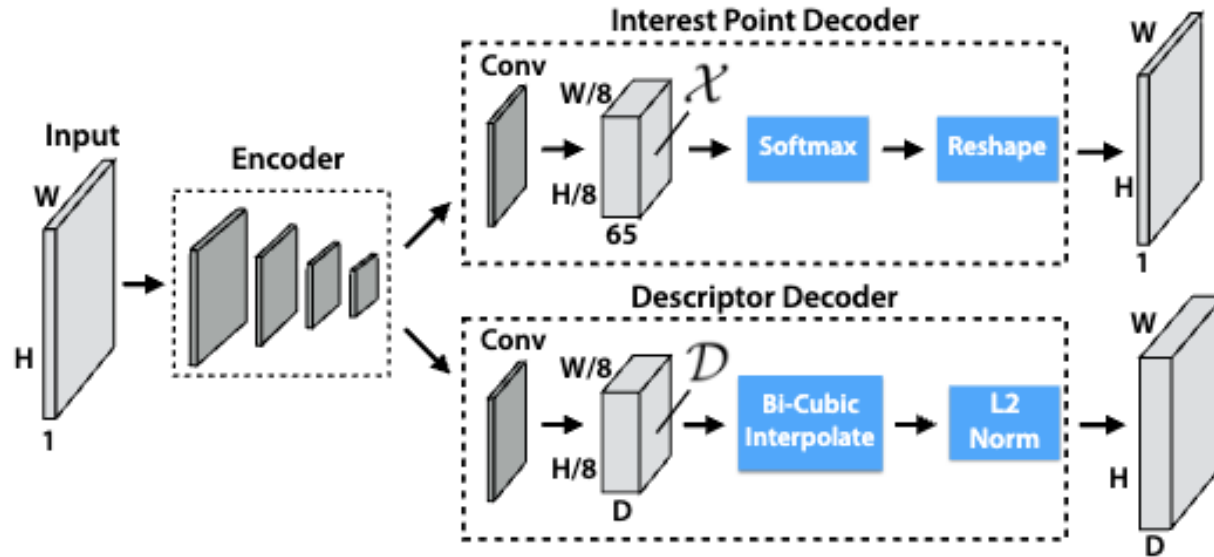
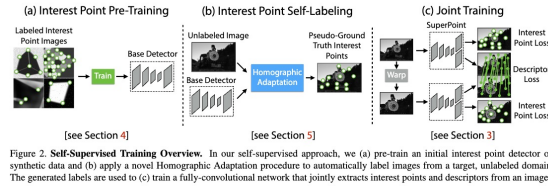
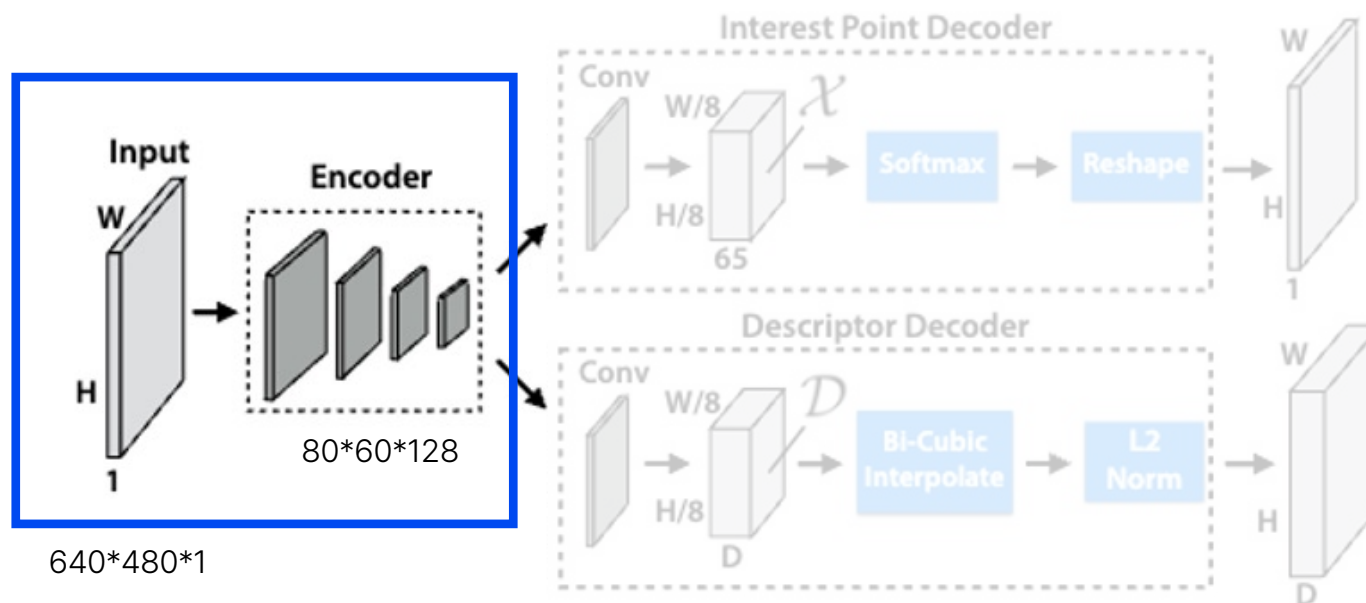


Figure 3. **SuperPoint Decoders.** Both decoders operate on a shared and spatially reduced representation of the input. To keep the model fast and easy to train, both decoders use non-learned upsampling to bring the representation back to $\mathbb{R}^{H \times W}$.

02 SuperPoint

2. SuperPoint Architecture

- Encoder : VGGNet 기반
- Height Weight가 1/8씩 축소되면서 압축된 텐서 80*60의 각 1픽셀은 원본의 64픽셀을 담당하게 됨.

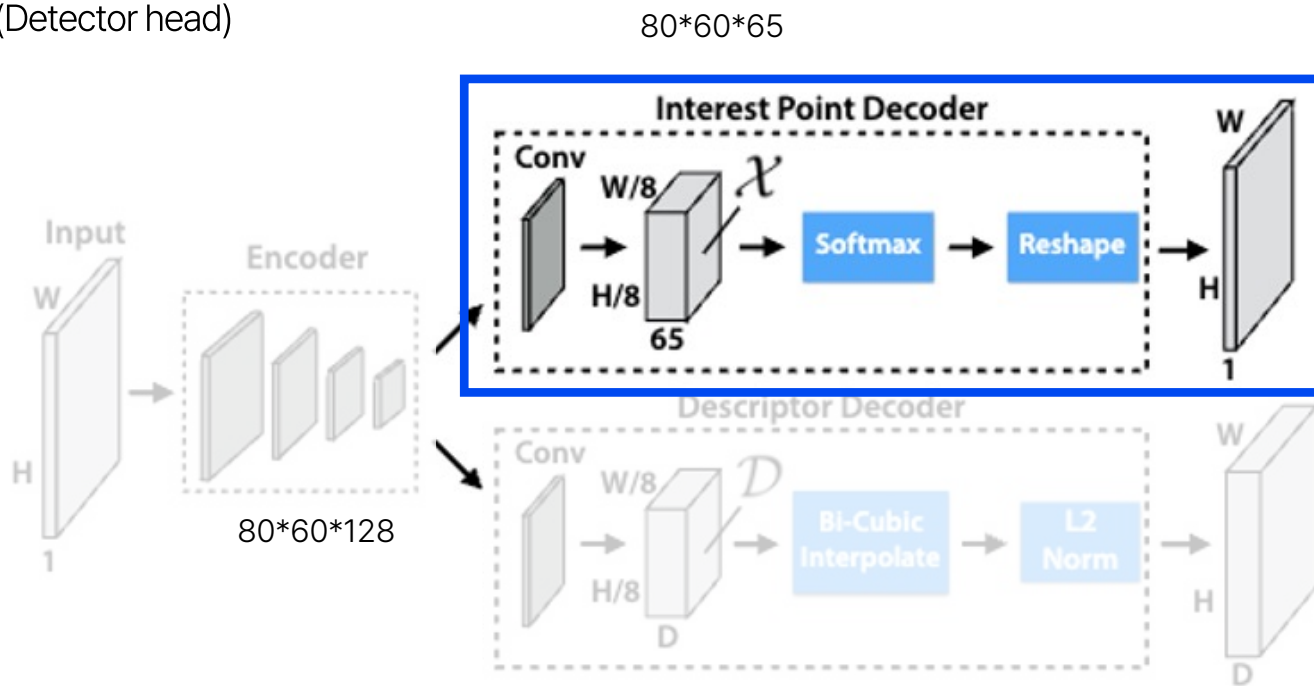


*Grayscale 변환 -> 기하학(코너, blob)적 특징은 밝기에서 나온다(색 정보X)

02 SuperPoint

2. SuperPoint Architecture

- Interest Point Decoder(Detector head)



1. Encoder에서 나온 텐서는 3*3conv, 1*1conv 통과해서 65채널의 텐서로 출력이 됨

- 1픽셀이 담당하는 $8 \times 8 = 64$ 픽셀을 의미하는 채널에 더하여, interest point가 없는 선택지인 1채널이 추가됨

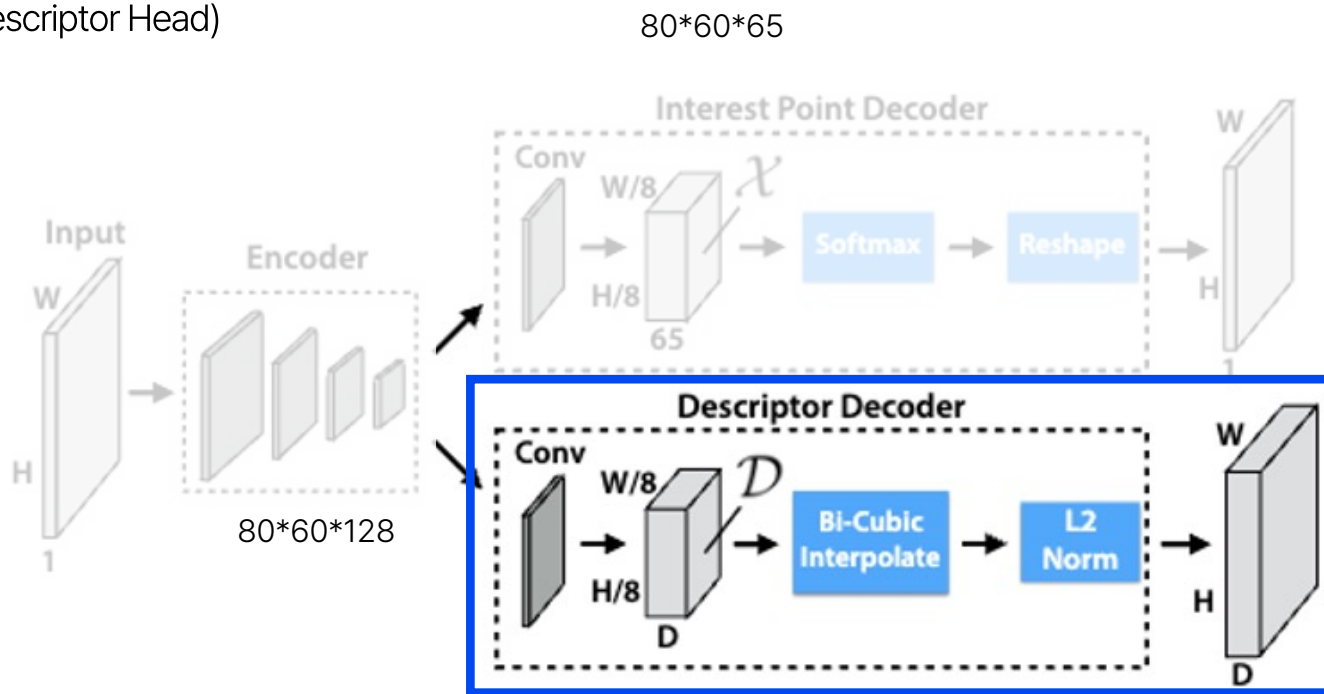
2. Softmax로 합이 1인 확률로 정규화

3. 원본 해상도로 reshape해서 heatmap 생성(interest point일 확률맵)

02 SuperPoint

2. SuperPoint Architecture

- Descriptor Decoder(Descriptor Head)



1. Detector가 찾은 점이 어떻게 생겼는지를 표현하는 벡터 D 차원의 벡터로 표현 : 이 벡터로 다른 이미지 B 와의 점을 매칭

- 채널의 의미가 다름 : 256채널은 특징 벡터의 차원

2. Bi-Cubic interpolate : 주변 픽셀을 고려한 보간방법

3. L2 norm : 길이를 1로 만들어서 패턴만 비교하기 위해.(크기 $\times$)

03 Train Stage

1. Synthetic Shapes

- 삼각형, 타원 등 단순 대규모 합성 데이터셋 생성
- 각 이미지에 homographic warp 사용하여 학습 데이터 증강

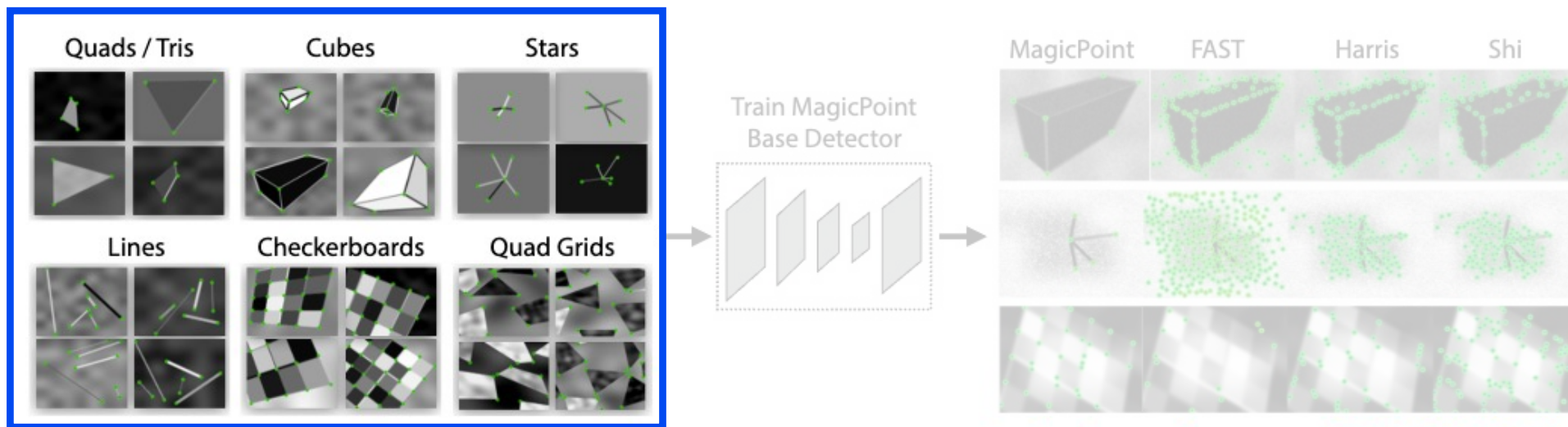
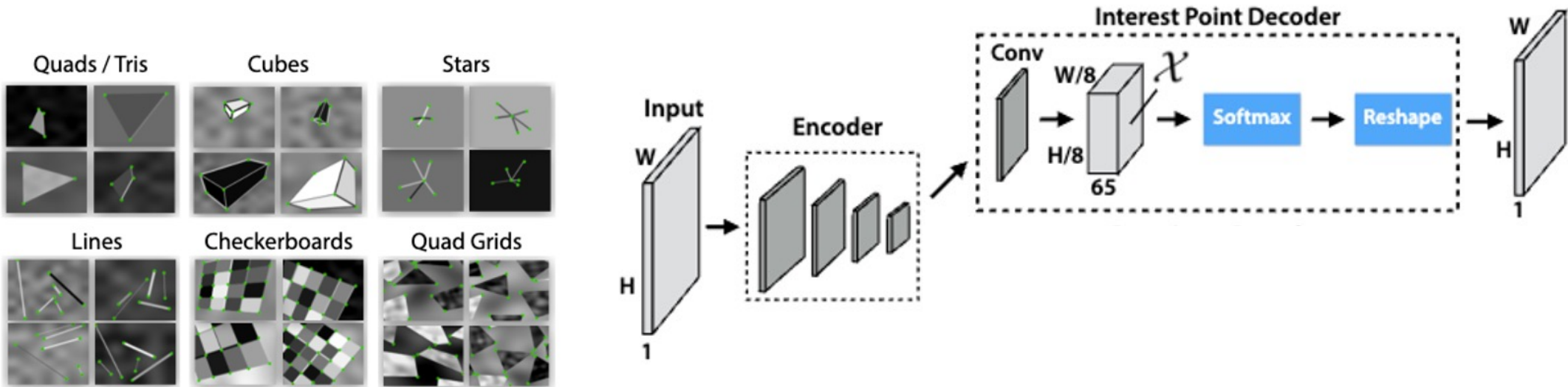


Figure 4. **Synthetic Pre-Training.** We use our Synthetic Shapes dataset consisting of rendered triangles, quadrilaterals, lines, cubes, checkerboards, and stars each with ground truth corner locations. The dataset is used to train the MagicPoint convolutional neural network, which is more robust to noise when compared to classical detectors.

03 Train Stage

2. MagicPoint

- MagicPoint : 아까 interest point detector 네트워크를 Synthetic Shapes 데이터로 학습시킨 모델을 의미



03 Train Stage

2. MagicPoint

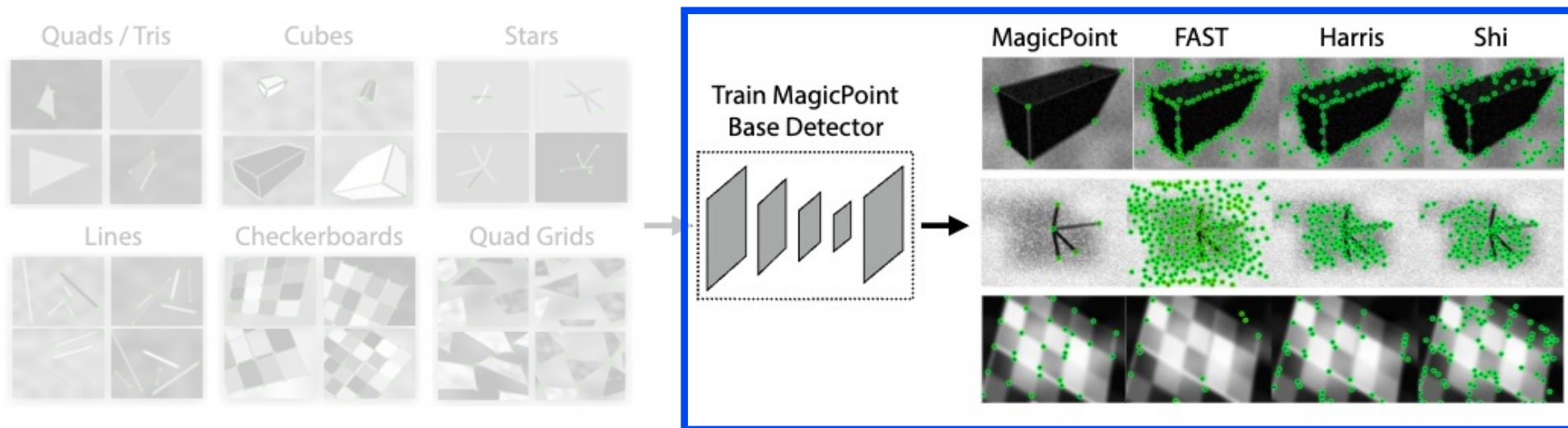


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- 기존의 전통적인 코너 검출 방식과 비교했을 때, 상당한 성능
- 전통적 검출기는 noise가 끼었을 때 성능저하
- Q. 그렇다면 실제 이미지로 일반화 할 수 있을까?

	MagicPoint	FAST	Harris	Shi
mAP no noise	0.979	0.405	0.678	0.686
mAP noise	0.971	0.061	0.213	0.157

*mAP : 검출한 점이 실제 정답 꼭지점과 얼마나 잘 맞는지, threshold를 바꿔가면 precision과 recall을 종합해 0~1로 요약한 점수

03 Train Stage

3. Homographic Adaptation

- MagicPoint는 실제 이미지에도 꽤 잘 작동하지만,, 자연 이미지에서 Repeatability 측면에서 성능저하
→ Homographic Adaptation : 여러 시점에서도 반복해서 잡혀야한다.

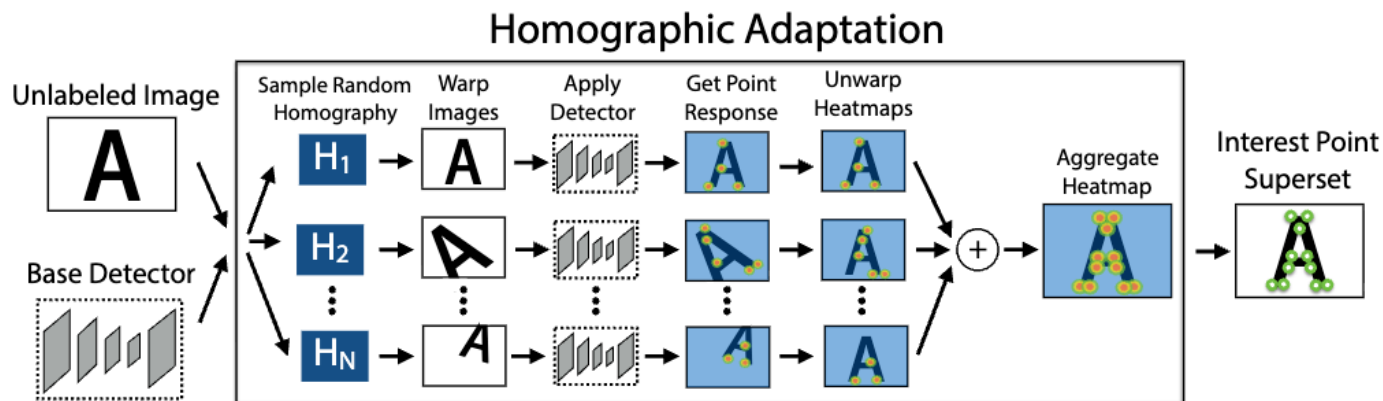


Figure 5. **Homographic Adaptation.** Homographic Adaptation is a form of self-supervision for boosting the geometric consistency of an interest point detector trained with convolutional neural networks. The entire procedure is mathematically defined in Equation 10.

- Homography란? 3*3 행렬로 표현되는 평면변환 : 이동/회전/원근 모두 포함
- 하나의 이미지를 다양하게 변형하여 예측하고 결과를 통합 → 믿을만한 정답(pseudo-label)생성
→ 일종의 앙상블 + Augmentation 효과

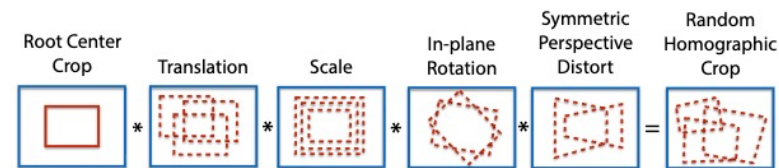


Figure 6. **Random Homography Generation.** We generate random homographies as the composition of less expressive, simple transformations.

03 Train Stage

3. Homographic Adaptation

- Homographic Adaptation을 진행할수록 detector 성능이 좋아지는 모습(정성)

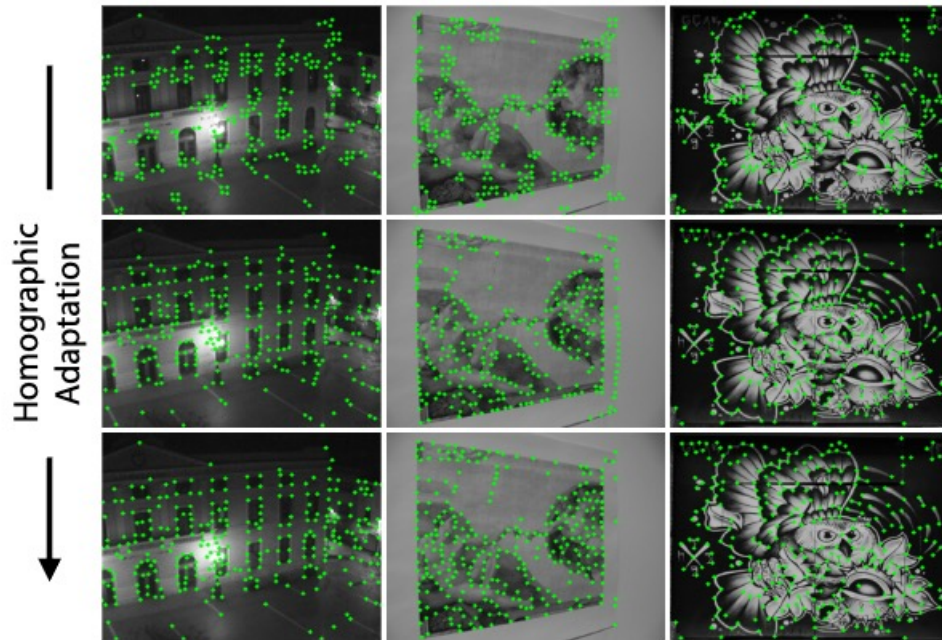


Figure 7. **Iterative Homographic Adaptation.** Top row: initial base detector (MagicPoint) struggles to find repeatable detections. Middle and bottom rows: further training with Homographic Adaption improves detector performance.

04 Experiments

1. 조명/시점에 따른 Repeatability 성능

< 실제 이미지에서 같은 점을 다시 찾는 비율(repeatability) 비교 >

*NMS(Non-maximum Suppression) : 주변부 중복제거 알고리즘

	조명	시점			
		57 Illumination Scenes		59 Viewpoint Scenes	
		NMS=4	NMS=8	NMS=4	NMS=8
MagicPoint + Homographic Adaptation =	<i>SuperPoint</i>	.652	.631	.503	.484
	<i>MagicPoint</i>	.575	.507	.322	.260
	<i>FAST</i>	.575	.472	.503	.404
	<i>Harris</i>	.620	.533	.556	.461
	<i>Shi</i>	.606	.511	.552	.453
	<i>Random</i>	.101	.103	.100	.104

Table 3. HPatches Detector Repeatability. SuperPoint is the most repeatable under illumination changes, competitive on view-point changes, and outperforms MagicPoint in all scenarios.

→ Homographic Adaptation을 적용했더니 시점변화에 robust해졌다

04 Experiments

2. 최종 성능

최종성능				점 찾기 성능		점 표현 성능	
Homography Estimation				Detector Metrics		Descriptor Metrics	
엄격	$\epsilon = 1$	$\epsilon = 3$	$\epsilon = 5$	Rep.	MLE	NN mAP	M. Score
SuperPoint	.310	.684	.829	.581	1.158	.821	.470
LIFT	.284	.598	.717	.449	1.102	.664	.315
SIFT	.424	.676	.759	.495	0.833	.694	.313
ORB	.150	.395	.538	.641	1.157	.735	.266

- * ϵ : 정답으로 치는 오차 기준
- *Rep. : repeatability
- *MLE : Mean Localization Error
- *NN mAP : 가장 가까운 descriptor끼리 올바른 짝을 얼마나 잘 찾는지
- *M. Score(Matching Score) : 올바르게 매칭된 비율

→ SuperPoint는 repeatability를 끌어올렸다

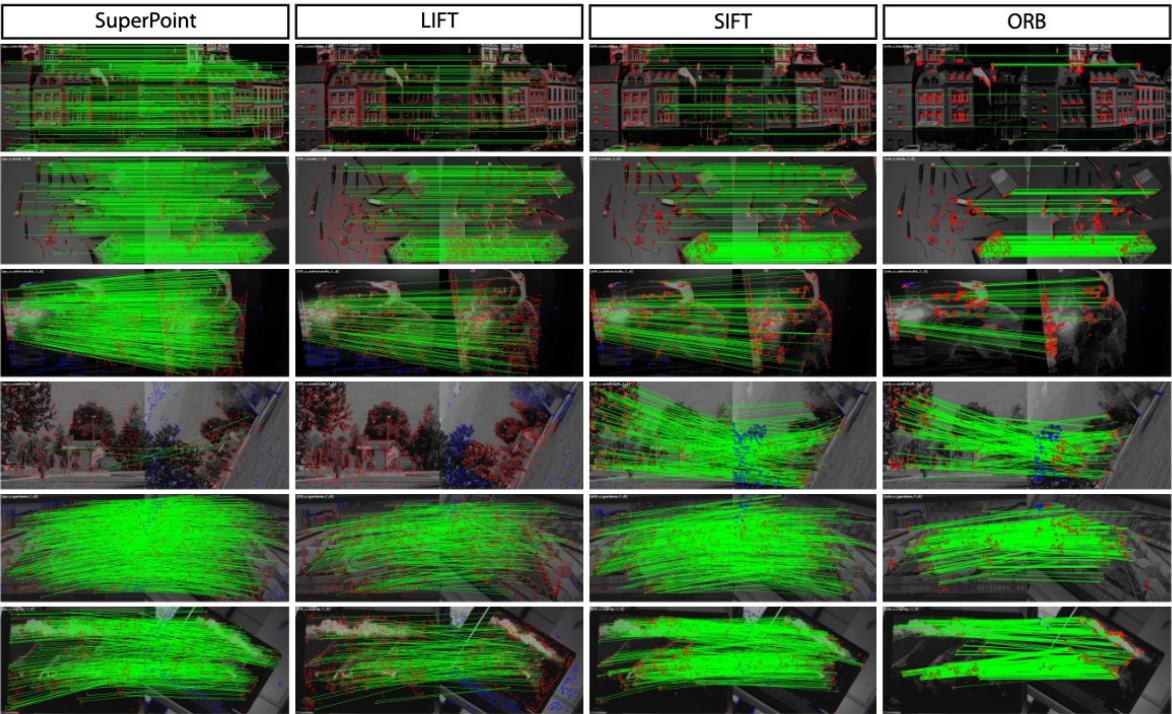


Figure 8. **Qualitative Results on HPatches.** The green lines show correct correspondences. SuperPoint tends to produce more dense and correct matches compared to LIFT, SIFT and ORB. While ORB has the highest average repeatability, the detections cluster together and generally do not result in more matches or more accurate homography estimates (see 4). Row 4: Failure case of SuperPoint and LIFT due to extreme in-plane rotation not seen in the training examples. See Appendix D for additional homography estimation example pairs.

05 Conclusion

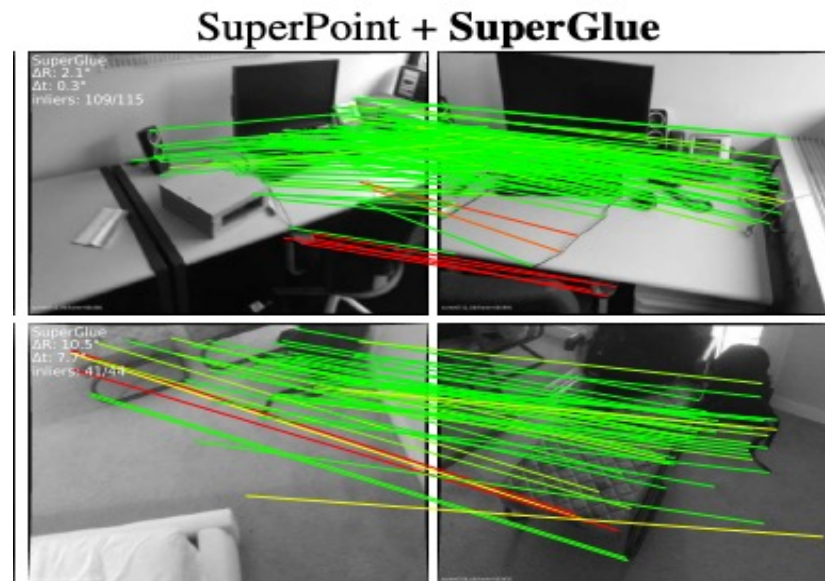
1. 정답이 없는 detection 문제를 self-supervised로 풀었다.

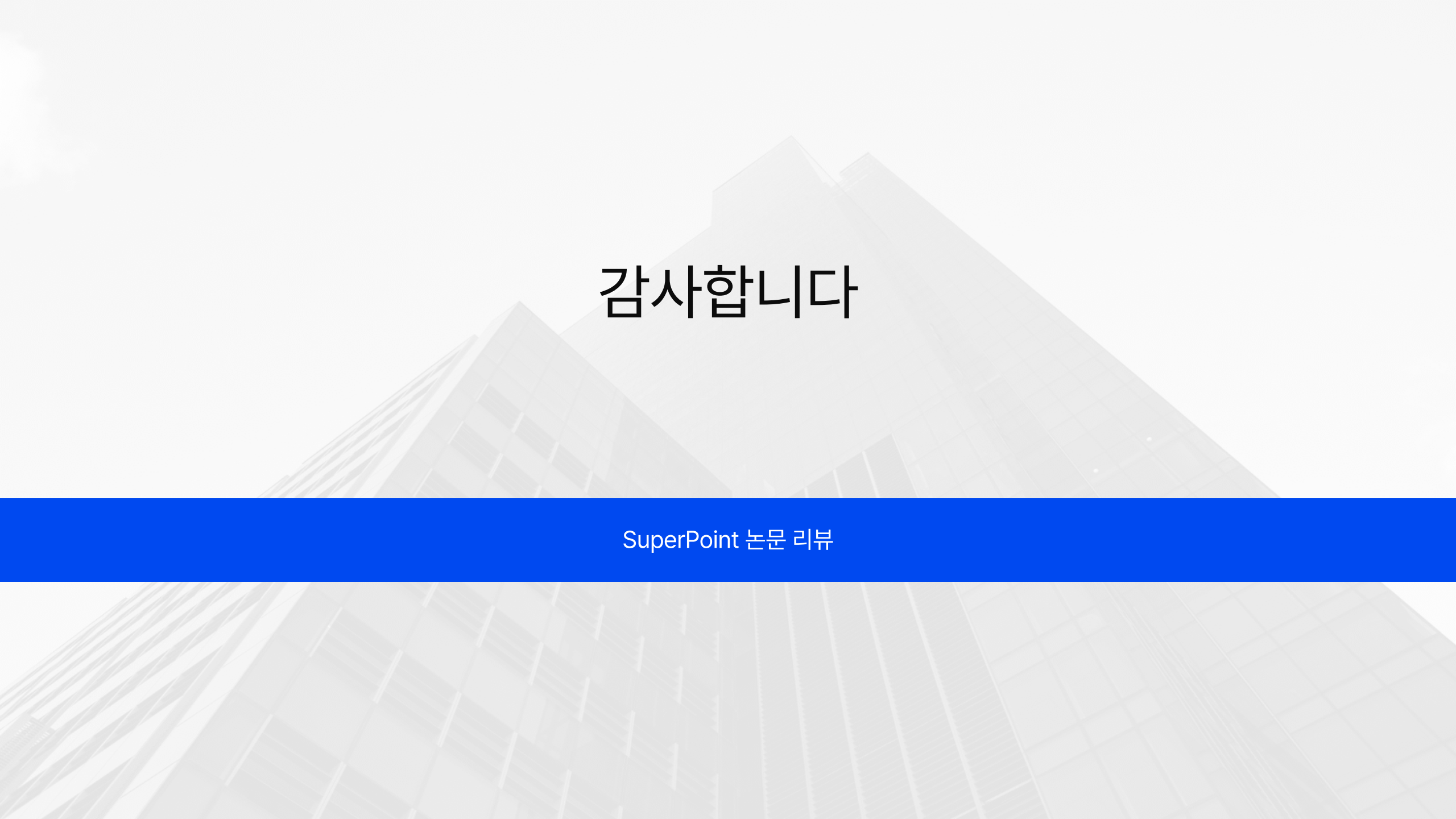
- 합성 도형(Synthetic shapes)로 기초 detector(MagicPoint)를 학습한 뒤, Homographic Adaptation으로 실제 이미지에서 스스로 pseudo-label(믿을만한 정답)을 만들어 학습.

2. SuperPoint는 “특징점을 어떻게 추출하는지”를 학습으로 풀어낸 연구 - keypoint detector

3. 후속 연구 SuperGlue

- SuperPoint가 뽑은 특징점들을 Matching 하는 후속 연구





감사합니다

SuperPoint 논문 리뷰